

Single-cell Transcriptomic Profiling of Matched Tumour and Normal Tissues from Colorectal Cancer Patients

1.0 Introduction

Colorectal cancer (CRC) is the third most common cancer and the second leading cause of cancer-related deaths worldwide (*Cancer Today*, n.d.). In 2022, 1.9 million new cases and 900,000 deaths were reported globally. CRC is strongly associated with advancing age, with adults aged 50 and above having the highest burden; however, recent studies have noted an increase in incidence among younger adults (Siegel et al., 2023). The standard CRC interventions (surgery, chemotherapy, radiotherapy, immunotherapy) remain unsatisfactory, with a drop in the 5-year survival rate among CRC patients from 91% at stage I to 12% at stage IV (Bazyari et al., 2023).

Single-cell RNA (scRNA) has profoundly revolutionized colorectal cancer studies by revealing cellular heterogeneity and providing a new understanding of the CRC microenvironment and its role in tumor progression. In this study, we aim to utilize scRNA transcriptomics to highlight the differentially expressed genes in colorectal tumors and normal tissues and explain their role in tumor progression.

2.0 Objectives

1. To justify quality control thresholds and report the number of cells and genes retained.
2. To cluster cell types and identify marker genes for each cluster.
3. To assign biological cell type identities.
4. To find differentially expressed genes between healthy and diseased states.

3.0 Methodology

3.1 Data source

scRNA sequencing data utilized in this study were obtained from the NCBI GEO database (accession: GSE294300), which includes matched samples (18 colorectal cancer samples and 18 adjacent-normal samples). The count matrix was downloaded using “wget”, along with the corresponding features and barcode data.

3.2 Quality control and normalization of data

We analysed the data using Seurat (v5.5.1) in R (v4.6.1). Initial quality control (QC) was performed, where we retained genes present in at least 3 cells to filter out extremely rare or uninformative genes and also retained individual cells that have at least 200 unique genes to filter low-quality cells (Wang et al., 2026a). A secondary QC was then done where we removed cells with fewer than 200 (indicative of empty droplets) or more than 6000 genes (indicative of doublets, which are two cells trapped inside a single droplet and sequenced

together) and those with more than 20% mitochondrial gene proportion (indicative of dying cells) (Wang et al., 2026a); (Tai et al., 2025). The count matrix was then normalized by dividing each cell's count by its total count, multiplied by a scale factor and then log-transformed to minimize the effects of expression outliers and to prevent high-abundance genes from dominating downstream analysis due to their high technical variability (Lytal et al., 2020).

3.3 Feature selection (Identification of highly variable genes (HVGs)), scaling, PCA, cell clustering and visualization

We selected 3000 HVGs and scaled each gene to remove differences in gene magnitude so that highly expressed genes do not dominate the principal component analysis (PCA) and ensure that all genes contribute comparably to the PCA. We then performed downstream analyses, including PCA (15 PCs), clustering using the Louvain algorithm, and visualized the cellular neighbours in 2D using the uniform manifold approximation projection (UMAP).

3.4 Cell annotation

To annotate cell populations, we applied singleR, a reference-based annotation method that compares the transcriptomic profile of each cluster with well-curated human cell atlases. To reduce computational memory usage and processing time while preserving biological diversity, a representative subset of cells was selected from each cluster (Up to 1,500 cells per cluster). We used three independent reference datasets (The Human Primary Cell Atlas (HPCA) for general human tissues, Blueprint Encode data for blood and immune cells, T-cells, B-cells, macrophages, etc., and Monaco Immune for distinguishing the CD4+ and CD8+ immune cells (Pasquini et al., 2021; Parums, 2024). Cell identities were predicted at the cluster level and mapped back to cells in the Seurat object. To further validate the stromal population, manually curated marker panels for cancer-associated fibroblasts (CAFs) and glial cells, derived from the literature, were evaluated using Seurat's module scoring approach (Wang et al., 2026b).

3.5 Differential expression analysis

Differential expression was computed using Seurat's DESeq2 integration to identify differentially expressed genes between normal and tumour tissues in cancer-associated fibroblasts, natural killer/CD8+ T cells, Schwann/glial cells, and epithelial cells. Statistical significance was set at p-adjusted <0.05 and a log2 fold-change (log2FC) of > 1 for upregulated genes and log2FC < -1 for downregulated genes.

4.0 Results

4.1 Single-cell transcriptomic map of colorectal cancer

After quality control, 312,200 cells and 38,850 genes were retained for downstream analysis. Dimensionality reduction, clustering, and UMAP revealed 31 distinct clusters belonging to 11 cell lineages based on canonical marker genes. We identified 7 immune cell lineages: macrophages, neutrophils, B, T/NK, plasma, dendritic and mast cells, and 4 non-immune lineages: fibroblasts, Schwann/glia, endothelial and epithelial cells.

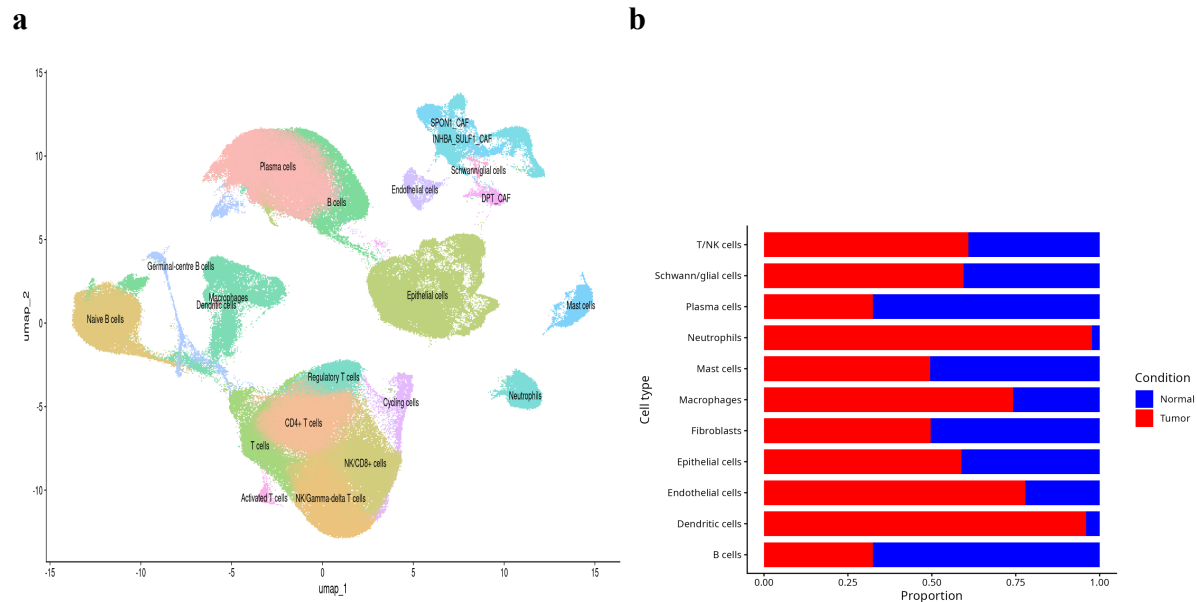
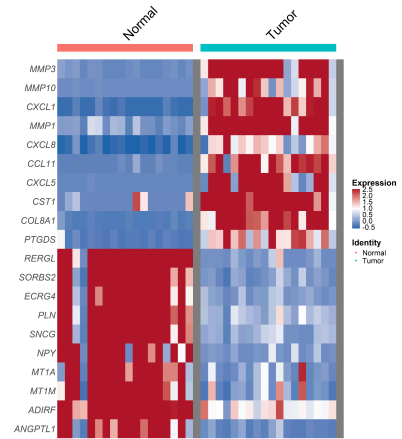
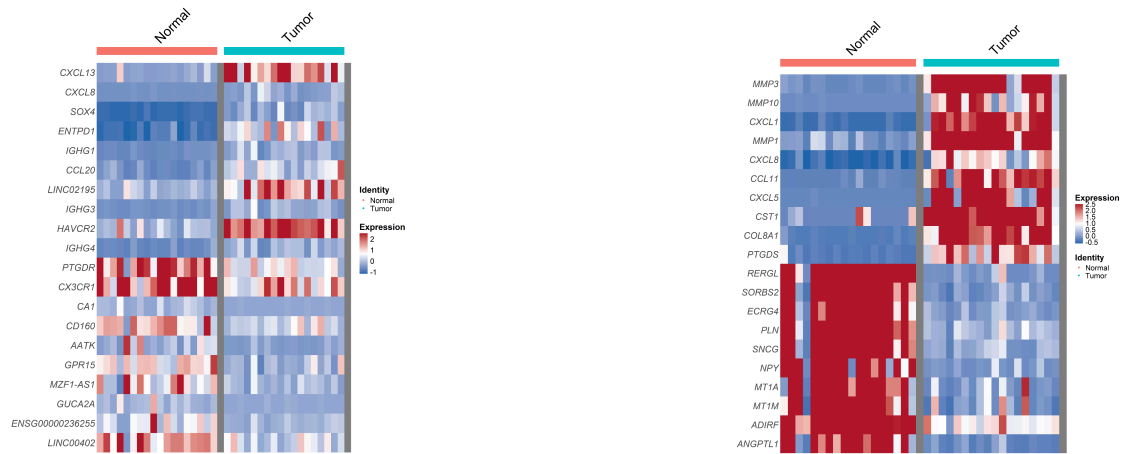


Figure 1. Single-cell transcriptomic landscape and cellular composition across tissue conditions. **a.** UMAP visualization of major cell types: Two-dimensional UMAP plot displaying the distinct clusters of isolated single cells. Individual cell clusters are colored and labeled according to their annotated cell identities, revealing major immune, stromal, and epithelial compartments. **b.** Cell type proportion across normal and tumor conditions: Stacked bar chart showing the relative proportions of each annotated cell type partitioned by sample condition. The horizontal axis represents the cell type proportion, with red bars denoting the fraction of cells derived from tumor tissue and blue bars representing normal tissue.

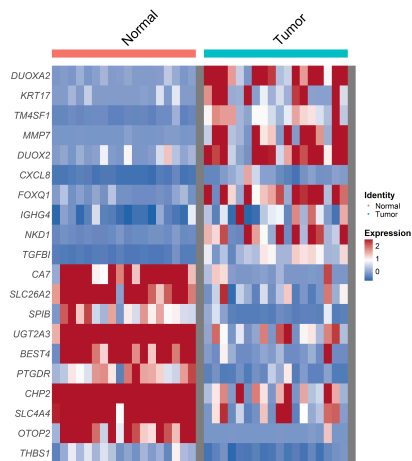
4.2 Differential Gene Expression (DGE) Analysis

Five cell lineages, T cells, Schwann/glia, fibroblasts, epithelial, and endothelial cells, were used for differential gene expression analysis.

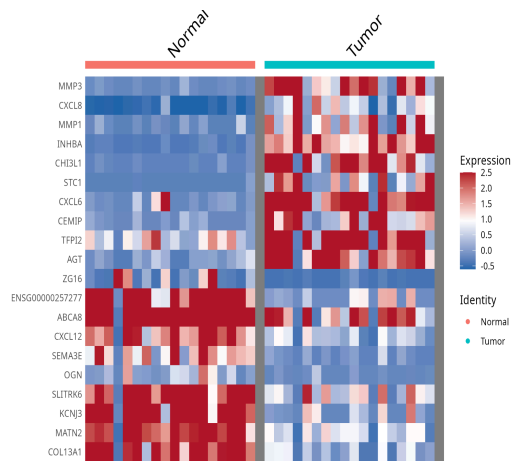




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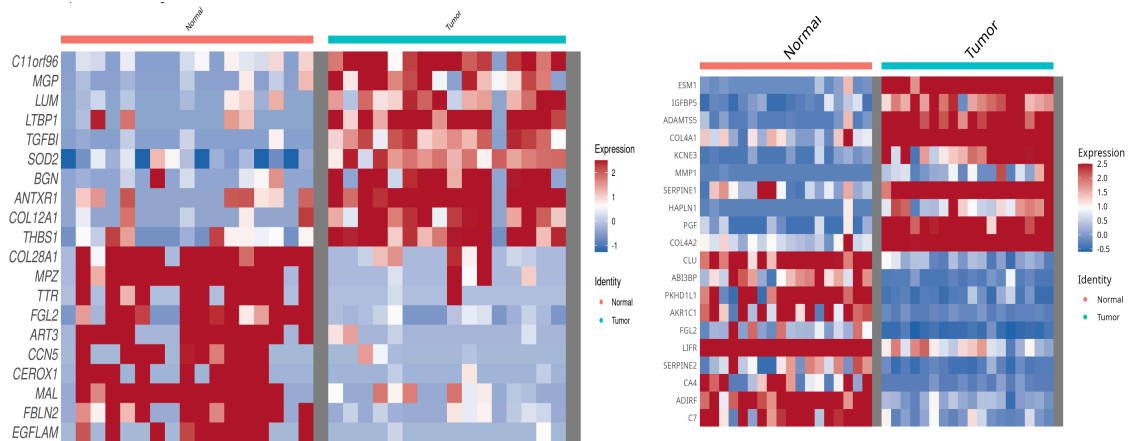


Figure 2. Heatmaps displaying the top differentially expressed marker genes between Normal (pink bar) and Tumor (teal bar) tissue samples across five key cell types. The relative expression levels are color-coded (red indicates high expression/upregulation; blue indicates low expression/downregulation). **a.** NK/CD8+ T cells: Differential gene expression profile highlighting key immune markers (e.g., *CXCL13*, *LINCO2195*, *HAVCR2*) upregulated in tumor; versus normal-associated markers like *CX3CR1*, *PTGDR*). **b.** *INHBA*+/*SULF8*+ CAFs: Matrix metalloproteinases and chemokines (e.g., *MMP3*, *MMP10*, *MMP1*, *CXCL1*, *CXCL8*) preferentially enriched in tumor-associated fibroblasts compared to normal tissue. **c.** Epithelial cells: Transcriptional reprogramming showing tumor-upregulated genes (e.g., *DUOXA2*, *KRT17*, *MMP7*) alongside markers suppressed relative to normal epithelial tissue (e.g., *CA7*, *SLC26A2*, *OTOP2*). **d.** *SPON1*+ CAFs: Stromal gene signatures displaying distinct upregulation of extracellular matrix (ECM) remodeling and signaling factors (e.g., *MMP3*, *CXCL8*, *INHBA*, *CHI3L1*) in tumor-derived *SPON1*+ CAFs. **e.** Schwann/Glial cells: Markedly distinct expression profiles revealing upregulation of stromal/ECM-related markers (*MGP*, *LUM*, *LTBP1*, *TGFB1*, *BGN*) in tumor tissue alongside loss of normal glial lineage identity genes (*MPZ*, *TTR*, *MAL*, *EGFLAM*). **f.** Endothelial cells have marked upregulation of *ESM1*, *IGFBP5* and *ADAMT55* genes.

5.0 Discussion

Natural killer (NK) cells are innate lymphocytes endowed with potent cytotoxic activity. The presence of tumor-associated NK cells has been correlated with better prognosis in several solid tumors including colorectal cancer (CRC) (Obino et al., 2025). In these cells, *CXCL13*, *LINCO2195*, *HAVCR2* genes are significantly overexpressed. Our findings are in agreement with (Nunez et al., 2020) who observed that C-X-C motif chemokine ligand 13 (*CXCL13*) treatment of CRC cell lines significantly increased cell proliferation and (Sakuma et al., 2024) who reported strong expression of Hepatitis A virus cellular receptor 2 (*HAVCR2*) with the infiltration of Dendritic cells (DCs) within the tumor microenvironment. Conversely, C-X3-C motif chemokine receptor 1 (*CX3CR1*) and Prostaglandin D2 Receptor (*PTGDR*)

were underexpressed in the CRC tumour tissues and overexpressed in the normal tissues. These findings were in agreement with (Gustafsson et al., 2007) who observed fivefold lower expression of PTGDR in CRC, compared to normal tissues as it inhibits cancer cell growth, proliferation, and migration. According to (Yue et al., 2022) silencing of *CX3CR1* and coculture with THP-1 cells further regulated macrophage polarization and promoted cell proliferation and migration of CRC cells.

Cancer-associated fibroblasts (CAFs) are key components of the tumour microenvironment, and they greatly shape the function and progression of cancer (Yang et al., 2023). We observed that Matrix metalloproteinases (*MMP*), (C-X-C motif) ligand (*CXCL*) and Inhibin beta-A (*INHBA*) genes were upregulated in CAFs. MMP genes facilitate tumour growth and metastasis in colorectal cancer (Liang et al., 2021). Additionally, Wang et al. reported an abundance of INHBA CAFs in colorectal tumour tissues (Wang et al., 2026c).

Tumour tissues had a higher proportion of endothelial cells in comparison to normal tissues. The Endothelial Cell-Specific Molecule 1 (ESM1) gene was significantly upregulated in tumour endothelial cells relative to those in normal tissues. This gene promotes angiogenesis (PubChem, n.d.), which is a key hallmark of cancer.

Epithelial cells are central to tumor progression, with a shift from functional homeostasis in the normal state to being remodelled in carcinogenesis. In healthy intestines, the epithelial cells play a major function in maintaining the wall lining homeostasis. However, in carcinogenesis, their transcription signatures have been shown to change, increasing their invasiveness by activating Epithelial-Mesenchymal Transition (EMT) (Xu et al., 2025). Various genes including Dual Oxidase Maturation Factor 2 (*DUOX2*), Dual Oxidase 2 (*DUOX2*), Keratin 17 (*KRT17*), Transmembrane 4 L Six Family Member 1 (*TM4SF1*), and *MMP7* were observed to be upregulated in tumor epithelial cells. These have been reported in CRC tissues, commonly associated with proliferation, EMT activation, apoptosis evasion and cell migration (Tang et al., 2020; Ujiie et al., 2021; Zhang et al., 2020; Zhou et al., 2023)

The gastrointestinal tract contains its own nervous system, known as the enteric nervous system (ENS), which is often referred to as "the second brain." The ENS consists of enteric neurons, enteric glial cells, and schwann cells associated with extrinsic peripheral nerves. As shown in our results, the genes such as *C11orf96*, *LUM*, *MGP*, *BGN*, *COL1A2*, *THBS1*, *TGFBI*, and *LTBP1* are upregulated in tumour tissues compared with normal tissues. *C11orf96* (Chromosome 11 open reading frame 96) is an emergency biomarker in colorectal cancer that is significantly upregulated during the transition from benign adenoma to malignant carcinoma suggesting a potential role in colorectal cancer progression. However,

its biological function and molecular mechanisms in CRC remain largely unexplored and require further experimental validation (Hauptman et al., 2018). *MGP* (Matrix Gla Protein) is an extracellular matrix protein that is overexpressed in CRC. MGP enriches intracellular free calcium concentration, which activates the Nk-kb signaling pathways (master regulator of inflammation, immune response and cell survival) to drive cancer growth. These findings identify *MGP* as a promising prognostic biomarker and potential therapeutic target for CRC (Li et al., 2020). *LUM* (Lumican) a small leucine-rich proteoglycan found in the tissue surrounding cells (extracellular matrix). *LUM* promotes tumour progression by enhancing immune escape, epithelial-to-mesenchymal transition (EMI), and metastasis through the miR200 signaling pathways that are involved in different aspects of cancer biology by targeting and repressing the expression of several key messenger RNAs (Chen & Zhang, 2017; Zang et al., 2021).

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